

Elasticity and Fluid Mechanics

How solids stretch and break, and how liquids and gases push, float and flow, from bones and cables to blood vessels and lungs.

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Until now our bodies have been rigid and our surroundings empty. Real materials deform under load, and engineers must know how much they stretch and when they fail. Much of the world is also fluid: water, air, oil, and blood. This chapter covers the mechanics of deformable solids (stress, strain, elastic moduli) and of fluids: pressure, buoyancy, and flow, with and without viscosity. These ideas underpin structural and mechanical engineering, hydraulics, aerodynamics, and a remarkable amount of physiology: blood pressure, circulation, breathing and drug delivery are all fluid mechanics.

YOU WILL BE ABLE TO

- Define stress and strain, and use Young's, bulk and shear moduli. Interpret stress–strain curves and failure.
- Calculate hydrostatic pressure, and distinguish gauge from absolute pressure.
- Apply Pascal's principle (hydraulics) and Archimedes' principle (buoyancy).
- Use the continuity equation and Bernoulli's equation for ideal flow.
- Apply Poiseuille's law to viscous flow in tubes, and recognise when flow becomes turbulent.
- Use surface tension and the law of Laplace for bubbles, alveoli and vessels.

THINK FIRST

A doctor measures blood pressure on the upper arm, held at the level of the heart. Would the reading be higher, lower, or the same if it were taken at the ankle of a standing patient? Why?

Much higher. The blood in the arteries forms a continuous column of fluid. A point about 1.3 m below the heart has an extra hydrostatic pressure of $\rho gh \approx 1060 \times 9.8 \times 1.3 \approx 13.5$ kPa, which is about 100 mmHg. A normal 120 mmHg systolic reading at the arm would read about 220 mmHg at the ankle. That is why blood pressure is always measured at heart level (Example 10.3).

§10.1

Stress, strain and elastic moduli

When forces act on a solid, it deforms. We describe the cause by the **stress** (force per unit area) and the effect by the **strain** (fractional deformation).

Tension and compression. A rod of length L_0 and cross-sectional area A , loaded along its axis by a force F :

$$\text{stress } \sigma = \frac{F}{A} \text{ (Pa)}, \quad \text{strain } \varepsilon = \frac{\Delta L}{L_0} \text{ (dimensionless)}.$$

For small strains, stress is proportional to strain (Hooke's law for materials):

$$\sigma = Y\varepsilon \implies \Delta L = \frac{FL_0}{YA}.$$

Y (often written E) is **Young's modulus**, a property of the material, not of the particular object. Comparing with a spring, $F = k\Delta L$, a rod behaves like a spring of stiffness $k = YA/L_0$.

Volume change. A pressure change Δp acting on all sides changes the volume by ΔV . The **bulk modulus** is

$$B = -\frac{\Delta p}{\Delta V/V}.$$

The minus sign makes B positive, since an increase in pressure reduces the volume. Its inverse, $1/B$, is the **compressibility**.

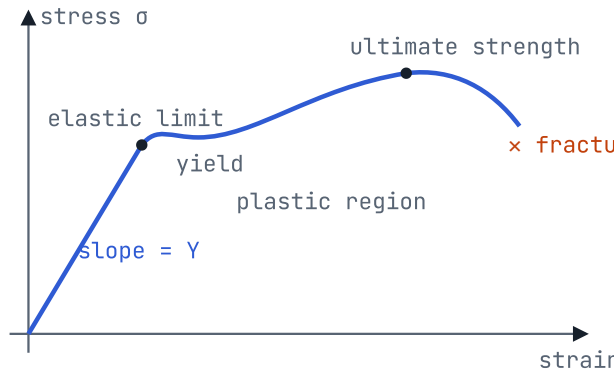
Shear. A tangential force F applied across a face of area A tilts the block by a small angle ϕ . The **shear modulus** is

$$S = \frac{F/A}{\phi}.$$

Fluids cannot sustain a static shear stress, so $S = 0$ for fluids. That is what makes them fluids.

MATERIAL	YOUNG'S MODULUS Y (GPA)	ULTIMATE TENSILE STRENGTH (MPA)
Steel (structural)	200	400–550
Aluminium	70	90–600
Concrete	30	2–5 (compressive: 20–40)
Cortical bone	15–20	130 (compressive: 170–200)
Tendon	1–2	50–100
Rubber	0.01–0.1	15
Carbon fibre composite	70–200	600–3500

Stress–strain behaviour and failure. Up to the *proportional limit*, stress is proportional to strain. Up to the *elastic limit*, the material returns to its original shape when unloaded. Beyond the **yield point**, ductile materials such as steel deform plastically and permanently. They then harden, reach the **ultimate tensile strength** (UTS), narrow (“neck”), and finally fracture. Brittle materials (glass, concrete, bone under fast loading) fail with little plastic deformation. Engineers design with a **safety factor**: the working stress is kept to a fraction (typically $\frac{1}{3}$ to $\frac{1}{5}$) of the yield stress.



A typical stress–strain curve for a ductile metal. The initial slope is Young's modulus. Beyond the elastic limit the deformation is permanent.

WORKED EXAMPLE 10.1

Stretching a steel wire

A steel wire 2.0 m long and 1.0 mm in diameter supports a 100 N load. Find the stress, the strain and the extension. Is it safe, given a yield stress of 250 MPa and a required safety factor of 3?

Solution

$$A = \pi(0.50 \times 10^{-3})^2 = 7.85 \times 10^{-7} \text{ m}^2, \quad \sigma = \frac{100}{7.85 \times 10^{-7}} = 1.27 \times 10^8 \text{ Pa} = 127 \text{ MPa}$$

$$\epsilon = \frac{\sigma}{Y} = \frac{1.27 \times 10^8}{2.0 \times 10^{11}} = 6.4 \times 10^{-4}, \quad \Delta L = \epsilon L_0 = 1.3 \text{ mm}.$$

Safety. The allowed working stress is $250/3 = 83$ MPa, so at 127 MPa the wire is **not** acceptable. It would need a diameter of at least $1.0\sqrt{127/83} = 1.24$ mm.

WORKED EXAMPLE 10.2

How strong is a thigh bone?

The shaft of an adult femur has about 4.0 cm² of cortical bone in cross-section and is about 0.45 m long. Take $Y = 17$ GPa and a compressive strength of 170 MPa. (a) How much does the femur shorten when a 700 N person stands on one leg? (b) What compressive force would fracture it?

Solution

(a) Shortening under 700 N:

$$\Delta L = \frac{FL_0}{YA} = \frac{700(0.45)}{(1.7 \times 10^{10})(4.0 \times 10^{-4})} = 4.6 \times 10^{-5} \text{ m} = 0.046 \text{ mm.}$$

(b) Fracture force:

$$F_{\max} = \sigma_{\max} A = (1.7 \times 10^8)(4.0 \times 10^{-4}) = 6.8 \times 10^4 \text{ N,}$$

about 100 times body weight.

Evaluate. Bone has a large static safety margin, yet the stiff-legged landing of Example 6.6 produced forces of tens of kilonewtons, the same order as this fracture load. In reality, bending (not pure compression) and fast impacts make bone fail at lower loads. Osteoporosis reduces both the cortical area and the bone density, cutting the strength further.

§10.2

Pressure in fluids

Pressure is the normal force per unit area, $p = F/A$, a scalar measured in pascals ($1 \text{ Pa} = 1 \text{ N/m}^2$). Other units remain common:

- $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa} = 760 \text{ mmHg}$
- $1 \text{ mmHg} = 133.3 \text{ Pa}$ (used for blood pressure)
- $1 \text{ bar} = 10^5 \text{ Pa}$
- $1 \text{ psi} = 6.89 \text{ kPa}$ (used for tyre pressure)

A fluid at rest pushes perpendicular to every surface it touches, and at any point the pressure is the same in all directions.

Variation of pressure with depth

Take a thin horizontal slab of fluid with area A and thickness dy , with y measured upward. The pressure below the slab must support both the pressure above it and its weight:

$$p(y)A - p(y + dy)A = \rho g A dy \implies \frac{dp}{dy} = -\rho g.$$

For an incompressible liquid, integrating gives the pressure at depth h below a surface where the

pressure is p_0 :

$$p = p_0 + \rho gh.$$

The pressure depends only on depth, not on the shape of the container (the *hydrostatic paradox*). Points at the same level in a connected body of fluid at rest are at the same pressure.

Gauge pressure is pressure above atmospheric, $p - p_{\text{atm}}$. Tyre gauges and blood-pressure cuffs read gauge pressure. **Absolute pressure** includes the atmosphere.

Measuring pressure. In a **barometer**, the column of mercury is held up by atmospheric pressure: $p_{\text{atm}} = \rho_{\text{Hg}}gh$, so $h = 760$ mm at standard pressure. A water barometer would need a column 10.3 m tall. An open-tube **manometer** measures gauge pressure as the difference in height between its two liquid columns.

WORKED EXAMPLE 10.3

Blood pressure from head to toe

In a standing adult, the head is about 0.40 m above the heart and the feet are about 1.3 m below it. The mean arterial pressure at heart level is 100 mmHg (gauge). Find the mean arterial pressure in the head and in the feet. The density of blood is 1060 kg/m^3 .

Solution

$$\rho g = 1060(9.8) = 1.04 \times 10^4 \text{ Pa/m} = 78 \text{ mmHg per metre.}$$

- Head: $\Delta p = -78(0.40) = -31 \text{ mmHg}$, so $p \approx 69 \text{ mmHg}$.
- Feet: $\Delta p = +78(1.3) = +101 \text{ mmHg}$, so $p \approx 200 \text{ mmHg}$.

Evaluate. Gravity redistributes pressure enormously. That is why ankles swell after long periods of standing, why you may feel faint when you stand up quickly (the pressure in the head drops before reflexes compensate), and why blood pressure is measured at heart level. A giraffe, whose head is about 2 m above its heart, needs a heart pressure of about 250 mmHg to supply its brain.

IN PRACTICE

Clinical hydrostatics. An IV bag must hang high enough that the pressure at the needle exceeds the patient's venous pressure (Problem P10.9). Raising the bag increases the flow rate. Central venous pressure is read in cm of water using a column referenced to the level of the heart. Divers gain about 1 atm of pressure for every 10 m of seawater, which compresses the air in their lungs and ears and dissolves extra nitrogen in their tissues. An ascent that is too fast lets that nitrogen form bubbles: decompression sickness, treated by re-pressurising the patient in a hyperbaric chamber.

§10.3

Pascal's principle and hydraulics

KEY IDEA

Pascal's principle. A change in pressure applied to an enclosed incompressible fluid is transmitted undiminished to every part of the fluid and to the walls of its container.

In a **hydraulic press**, a small force F_1 on a small piston of area A_1 creates a pressure F_1/A_1 , which acts on a large piston of area A_2 :

$$F_2 = F_1 \frac{A_2}{A_1}.$$

Energy is conserved: the volume pushed in equals the volume pushed out, so the small piston must move farther, $d_1 = d_2 A_2 / A_1$, and $F_1 d_1 = F_2 d_2$. Car brakes, excavators, aircraft control surfaces and hospital beds all use hydraulics.

§10.4

Buoyancy: Archimedes' principle

KEY IDEA

Archimedes' principle. A body wholly or partly immersed in a fluid feels an upward **buoyant force** equal to the weight of the fluid it displaces:

$$F_B = \rho_{\text{fluid}} V_{\text{displaced}} g.$$

The buoyant force comes from the pressure difference: the fluid pushes harder on the bottom of the body, which is deeper, than on the top.

- A body **sinks** if its average density exceeds the fluid's density.
- It **floats** if its average density is less. In equilibrium it displaces its own weight of fluid, so the fraction submerged is $\rho_{\text{body}}/\rho_{\text{fluid}}$.

WORKED EXAMPLE 10.4

Measuring body fat by underwater weighing

A person of mass 80.0 kg has an apparent mass of 3.6 kg when weighed fully submerged in water ($\rho_w = 1000 \text{ kg/m}^3$) after breathing out as far as possible. (a) Find the body's volume and average density. (b) Estimate the percentage of body fat using the Siri equation,

$$\% \text{fat} = \frac{495}{\rho \text{ (g/cm}^3\text{)}} - 450. \text{ Ignore the air left in the lungs for now.}$$

Solution

(a) The apparent loss of weight equals the buoyant force:

$$\rho_w V g = (80.0 - 3.6)g \Rightarrow V = \frac{76.4}{1000} = 0.0764 \text{ m}^3.$$

$$\rho = \frac{80.0}{0.0764} = 1047 \text{ kg/m}^3 = 1.047 \text{ g/cm}^3.$$

(b) Siri equation:

$$\% \text{fat} = \frac{495}{1.047} - 450 = 472.8 - 450 = 23\%.$$

Evaluate. The method works because fat (0.90 g/cm^3) is less dense than lean tissue (1.10 g/cm^3). It is very sensitive to errors in volume, and in particular to the air that remains in

the lungs. Problem P10.14 shows that ignoring about 1 L of residual air overestimates body fat by several percentage points.

§10.5

Fluids in motion: continuity

We first model an **ideal fluid**: incompressible, non-viscous, and in **steady** flow, so that the velocity at each point does not change with time. The path a fluid particle follows is a **streamline**.

Streamlines never cross, and where they crowd together the fluid moves faster.

Volume flow rate is $Q = Av$, measured in m^3/s . In steady incompressible flow, no fluid piles up anywhere, so the same volume must pass every cross-section each second:

KEY IDEA

Continuity equation: $A_1v_1 = A_2v_2$. Where the pipe narrows, the fluid speeds up. If a pipe branches into several, the *total* cross-sectional area is what counts: $Q = \sum A_i v_i$.

WORKED EXAMPLE 10.5

From the aorta to the capillaries

Blood leaves the heart through the aorta (cross-sectional area 3.0 cm^2) at an average speed of 30 cm/s . In the capillaries it moves at about 0.50 mm/s . (a) Find the total cross-sectional area of all the capillaries. (b) Each capillary has a radius of about $4 \text{ }\mu\text{m}$. Estimate how many capillaries are open at any time.

Solution

(a) By continuity, the total flow is the same everywhere:

$$A_{\text{cap}} = \frac{A_{\text{aorta}} v_{\text{aorta}}}{v_{\text{cap}}} = \frac{(3.0 \times 10^{-4})(0.30)}{5.0 \times 10^{-4}} = 0.18 \text{ m}^2.$$

(b) Each capillary has area $\pi(4 \times 10^{-6})^2 = 5.0 \times 10^{-11} \text{ m}^2$, so

$$N \approx \frac{0.18}{5.0 \times 10^{-11}} \approx 4 \times 10^9.$$

Evaluate. Each capillary is tiny, but there are billions of them, with a total area some 600 times that of the aorta, so the blood slows right down. That slow flow gives oxygen and

nutrients time to diffuse across the capillary walls, and it is no accident of design.

§10.6

Bernoulli's equation

Apply the work–energy theorem to a parcel of ideal fluid moving along a streamline. The pressure forces from the surrounding fluid do work on it, gravity does work on it, and the result is a change in its kinetic energy. Per unit volume, this gives:

KEY IDEA

Bernoulli's equation. Along a streamline in steady, incompressible, non-viscous flow,

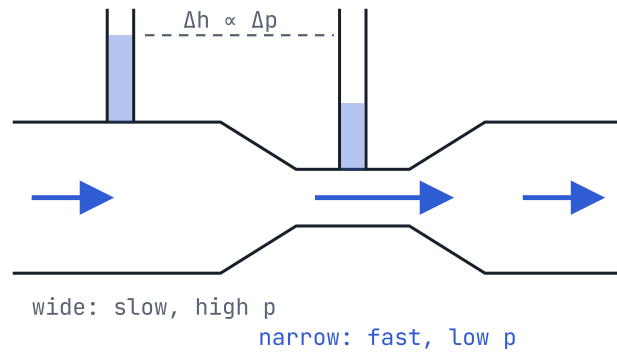
$$p + \frac{1}{2}\rho v^2 + \rho gy = \text{constant}.$$

Each term is an energy per unit volume. Where the fluid moves faster, at the same height, its pressure is **lower**.

Derivation sketch. In a time dt , a volume dV enters a section of pipe at height y_1 with pressure p_1 and speed v_1 , and the same volume leaves at height y_2 with p_2 and v_2 . The net work done by pressure is $(p_1 - p_2)dV$. The work done by gravity is $-\rho dV g(y_2 - y_1)$. The change in kinetic energy is $\frac{1}{2}\rho dV(v_2^2 - v_1^2)$. Setting the total work equal to the change in kinetic energy and rearranging gives Bernoulli's equation.

Special cases:

- $v = 0$ everywhere: $p + \rho gy$ is constant, which is hydrostatics again. ✓
- **Torricelli's law.** Fluid flows from a small hole at depth h below the open surface of a large tank. Both points are at atmospheric pressure, so $v = \sqrt{2gh}$, the same speed as a free fall through h .
- **Venturi meter.** In a constriction the fluid speeds up and its pressure drops. Measuring the pressure drop gives the flow rate.



A Venturi tube. Fluid speeds up through the constriction, so its pressure falls, shown by the lower liquid column. The pressure difference measures the flow rate.

WORKED EXAMPLE 10.6

A narrowed artery

An atherosclerotic plaque narrows an artery to one-third of its normal cross-sectional area. Upstream, blood flows at 0.50 m/s. Treating the blood as an ideal fluid, find the speed in the narrowed segment and the drop in pressure there.

Solution

Continuity: $v_2 = v_1(A_1/A_2) = 3(0.50) = 1.5$ m/s.

Bernoulli, horizontal flow:

$$p_1 - p_2 = \frac{1}{2}\rho(v_2^2 - v_1^2) = \frac{1}{2}(1060)(2.25 - 0.25) = 1060 \text{ Pa} \approx 8 \text{ mmHg}.$$

Evaluate. The pressure inside the narrowing falls. If the pressure outside the vessel exceeds the reduced internal pressure, the artery can partly collapse, briefly stopping the flow. The pressure then builds, the vessel reopens, and the cycle repeats. This *flutter* can damage the plaque. The fast jet beyond the narrowing is also often turbulent, producing a *bruit*, a sound a doctor can hear with a stethoscope. Doppler ultrasound measures the jet speed, and the simplified Bernoulli relation $\Delta p \approx 4v^2$ (with Δp in mmHg and v in m/s) is used clinically to estimate pressure drops across narrowed heart valves.

IN PRACTICE

Bernoulli in engineering and nature. Carburettors and atomisers draw fuel or liquid into a fast, low-pressure air stream. Pitot tubes on aircraft measure airspeed from the difference between the stagnation pressure ($p + \frac{1}{2}\rho v^2$) and the static pressure. Wind blowing across a burrow's raised exit lowers the pressure there and ventilates the burrow, which prairie dogs exploit. Aircraft lift is often "explained" by Bernoulli alone, but the full explanation needs the wing to deflect air downward: Bernoulli's equation and Newton's third law describe the same flow from two viewpoints.

§10.7

Viscous flow: Poiseuille's law

Real fluids are **viscous**: adjacent layers moving at different speeds exert a frictional shear force on each other. For a fluid between two plates, the shear stress is proportional to the velocity gradient:

$$\frac{F}{A} = \eta \frac{dv}{dy}.$$

The constant η is the **viscosity**, measured in Pa s. For water at 20 °C, $\eta = 1.0 \times 10^{-3}$ Pa s. For blood it is about $3\text{--}4 \times 10^{-3}$ Pa s. Honey reaches about 10 Pa s.

In a long straight tube, viscosity makes the flow fastest at the centre and zero at the wall, with a parabolic velocity profile. Integrating that profile across the tube gives **Poiseuille's law**, the flow rate through a tube of radius r and length L driven by a pressure difference Δp :

$$Q = \frac{\pi r^4 \Delta p}{8\eta L}.$$

By analogy with Ohm's law ($I = V/R$, Chapter 17), we write $Q = \Delta p/R_{\text{flow}}$, with a **flow resistance**

$$R_{\text{flow}} = \frac{8\eta L}{\pi r^4}.$$

The extraordinary r^4 dependence dominates everything. Halving the radius increases the resistance **sixteen-fold**.

WORKED EXAMPLE 10.7

Vasoconstriction and an IV needle

(a) The body controls blood flow to organs mainly by contracting the smooth muscle around arterioles. By what factor does the flow drop, at fixed pressure, if an arteriole's radius decreases by 20%? (b) Saline ($\eta = 1.0 \times 10^{-3}$ Pa s) flows through a needle of internal radius 0.20 mm and length 3.0 cm, driven by a pressure difference of 8.0 kPa. Find the flow rate in mL/h.

Solution

(a) $Q \propto r^4$, so $Q'/Q = 0.80^4 = 0.41$. The flow drops to 41% of its original value. A small change in radius gives very powerful control of flow.

(b) Poiseuille's law:

$$Q = \frac{\pi(2.0 \times 10^{-4})^4(8.0 \times 10^3)}{8(1.0 \times 10^{-3})(0.030)} = \frac{\pi(1.6 \times 10^{-15})(8.0 \times 10^3)}{2.4 \times 10^{-4}} = 1.7 \times 10^{-7} \text{ m}^3/\text{s}.$$

Since $1 \text{ m}^3 = 10^6 \text{ mL}$, this is $0.17 \text{ mL/s} \approx 600 \text{ mL/h}$.

Evaluate. For rapid transfusion, clinicians choose short, wide-bore cannulas. A cannula with twice the radius delivers 16 times the flow, while a longer one delivers proportionally less. In practice the tubing and the blood's higher viscosity also matter, but the r^4 law is why cannula gauge is the critical choice in trauma care.

Turbulence. Poiseuille's law assumes smooth, laminar flow. When the Reynolds number $Re = \rho v D / \eta$ exceeds about 2000 (Chapter 1), the flow becomes turbulent and the flow resistance rises sharply. Turbulence explains the sounds heard when measuring blood pressure. As the cuff is released, blood squirts through the partly compressed artery in turbulent jets, producing the *Korotkoff sounds*. The pressure at which they begin is the systolic pressure, and the pressure at which they stop is the diastolic pressure.

§10.8

Surface tension and the law of Laplace

Molecules at a liquid's surface have fewer neighbours to attract them, so the surface behaves like a stretched membrane. The **surface tension** γ is the force per unit length along the surface, in N/m. For water at 20 °C, $\gamma = 0.073 \text{ N/m}$.

A curved surface therefore has a higher pressure on its concave side. This is the **law of Laplace**:

- spherical droplet or gas bubble in a liquid: $\Delta p = \frac{2\gamma}{r}$;
- soap bubble, which has two surfaces: $\Delta p = \frac{4\gamma}{r}$;
- cylindrical vessel, with wall tension T per unit length: $T = \Delta p r$.

Surface tension also drives **capillary rise**: a liquid that wets a thin tube climbs to a height $h = 2\gamma \cos \theta / (\rho g r)$. This is one of the ways water rises in soil.

WORKED EXAMPLE 10.8

Why lungs need surfactant

Model an alveolus as a sphere of radius 0.10 mm lined with a thin film of fluid. (a) With the surface tension of water (0.070 N/m), what pressure difference would be needed to keep it inflated? (b) Pulmonary surfactant lowers the surface tension to about 0.005–0.025 N/m. Repeat the calculation.

Solution

(a) $\Delta p = 2\gamma/r = 2(0.070)/(1.0 \times 10^{-4}) = 1400 \text{ Pa} \approx 10.5 \text{ mmHg}$.

(b) With $\gamma = 0.005\text{--}0.025 \text{ N/m}$: $\Delta p = 100\text{--}500 \text{ Pa}$, about 0.75–3.8 mmHg.

Evaluate. Without surfactant, breathing would require far more effort, and small alveoli, which by Laplace's law need *higher* pressure, would empty into larger ones and collapse (Problem P10.18). Premature babies often lack surfactant and develop respiratory distress syndrome. Treatment with artificial surfactant, delivered into the lungs, was a major advance in neonatal care.

CHAPTER SUMMARY

- Stress $\sigma = F/A$ and strain $\epsilon = \Delta L/L_0$, with $\sigma = Y\epsilon$, so $\Delta L = FL_0/YA$. Bulk modulus $B = -\Delta p/(\Delta V/V)$. Shear modulus $S = (F/A)/\phi$.
- Pressure $p = F/A$. Hydrostatics: $p = p_0 + \rho gh$. Gauge pressure = absolute – atmospheric.
- Pascal: pressure is transmitted undiminished, so hydraulic machines give $F_2 = F_1 A_2/A_1$.
- Archimedes: $F_B = \rho_{\text{fluid}} V_{\text{disp}} g$. A floating body has fraction $\rho_{\text{body}}/\rho_{\text{fluid}}$ submerged.
- Continuity: $A_1 v_1 = A_2 v_2$. Bernoulli: $p + \frac{1}{2} \rho v^2 + \rho gy$ is constant along a streamline. Torricelli: $v = \sqrt{2gh}$.
- Viscous flow (Poiseuille): $Q = \pi r^4 \Delta p / (8\eta L)$, so $R_{\text{flow}} \propto r^{-4}$. Turbulence above $Re \approx 2000$.
- Laplace: $\Delta p = 2\gamma/r$ for a droplet or alveolus, $4\gamma/r$ for a soap bubble, and $T = \Delta p r$ for a vessel wall.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P10.1

Even with a perfect vacuum pump at the top, water cannot be sucked up a vertical pipe higher than about 10 m. Why not? What limits the height?

P10.2

A rowing boat carrying a heavy iron anchor floats in a swimming pool. The anchor is thrown overboard and sinks to the bottom. Does the water level in the pool rise, fall or stay the same? Explain.

P10.3

Capillaries are far narrower than the aorta, yet blood flows much more slowly in them. Doesn't the continuity equation say that narrower tubes give faster flow? Resolve the apparent paradox.

P10.4

Why must a blood-pressure cuff be at the level of the heart? What error results if the patient's arm hangs down by their side with the cuff 30 cm below heart level?

LEVEL B – STANDARD PROBLEMS**P10.5**

A 1000 kg lift accelerates upward at up to 1.5 m/s^2 . It hangs from a single steel cable with a yield stress of 250 MPa, and a safety factor of 5 is required. Find the minimum cable diameter.

P10.6

The tibia has about 3.0 cm^2 of cortical bone in cross-section, with a compressive strength of 170 MPa. (a) What force would fracture it in pure compression? (b) Compare this with the forces found in Example 6.6 for a stiff-legged landing from 1.0 m, shared between two legs, and comment on the risk.

P10.7

In a hydraulic car lift, the small piston has an area of 5.0 cm^2 and the large piston 500 cm^2 . (a) What force on the small piston lifts a 1500 kg car? (b) How far must the small piston move in total to raise the car by 10 cm?

P10.8

A breath-hold diver fills her lungs to 6.0 L at the surface and dives to 30 m in seawater ($\rho = 1025 \text{ kg/m}^3$). (a) Find the absolute pressure at 30 m. (b) Assuming the air in her lungs stays at constant temperature (so pV is constant), find her lung volume at that depth. (c) A scuba diver at 30 m breathes air at the surrounding pressure. Explain why holding the breath while ascending is dangerous.

P10.9

An IV bag delivers saline into a vein where the gauge pressure is 15 mmHg. (a) What is the minimum height of the fluid surface in the bag above the needle for any flow at all? (b) If the bag is hung 1.0 m above the needle, what pressure difference drives the flow (in mmHg)?

P10.10

A cube of wood 0.20 m on a side, with density **600 kg/m³**, floats in water. (a) What depth of the cube is submerged? (b) What mass placed on top of the cube will just submerge it completely?

P10.11

Water flows at 2.0 m/s through a horizontal main of diameter 10 cm, at a gauge pressure of 200 kPa. The pipe then narrows to 5.0 cm in diameter and rises by 5.0 m. Find the speed and the gauge pressure in the narrow, raised section.

P10.12

An arteriole constricts so that its radius decreases by 10%. (a) By what percentage does its flow resistance increase? (b) If the pressure difference across it stays the same, by what percentage does the flow fall? (c) What pressure difference would be needed to restore the original flow?

LEVEL C – CHALLENGE PROBLEMS**P10.13**

Dam engineering. A dam holds back water to a depth **H** along a straight face of width **W** . (a) By integrating the pressure over the face, show that the total horizontal force of the water is **$F = \frac{1}{2}\rho gWH^2$** . (b) Show that this force acts at a height **$H/3$** above the base. (c) Evaluate **F** for **$H = 50$ m** and **$W = 200$ m**. Explain why dams are much thicker at the bottom.

P10.14

Body fat, done properly. A 75.0 kg subject's apparent mass when fully submerged is 2.5 kg. The residual air left in the lungs after a full exhalation is measured as 1.2 L. Take the water's density as **1000 kg/m³**. (a) Find the body's true average density, correcting for the residual air (the air's mass is negligible). (b) Use the Siri equation to find the body-fat percentage. (c) Repeat without the correction, and comment on the size of the error.

P10.15

Draining a tank. A cylindrical tank with cross-sectional area A is filled with water to height h_0 . It drains through a small hole of area a in the bottom. (a) Use Torricelli's law and continuity to show that $dh/dt = -(a/A)\sqrt{2gh}$. (b) Show that the tank empties in a time $t = (A/a)\sqrt{2h_0/g}$. (c) Evaluate this for $A = 1.0 \text{ m}^2$, a hole of radius 1.0 cm, and $h_0 = 2.0 \text{ m}$. Why does the second half of the water take longer to drain than the first?

P10.16

Stenosis. A plaque reduces an artery's radius to half its normal value over a short length. Upstream, the artery has a diameter of 8.0 mm and the blood speed is 0.40 m/s. Take $\rho = 1060 \text{ kg/m}^3$ and $\eta = 3.5 \times 10^{-3} \text{ Pa s}$. (a) Find the speed in the narrowed section and the Bernoulli pressure drop there. (b) By what factor does the viscous (Poiseuille) resistance of the narrowed segment exceed that of a normal segment of the same length? (c) By what factor does the Reynolds number change in the narrowed section? Comment on whether a bruit (turbulent sound) is likely.

P10.17

The atmosphere. For an isothermal atmosphere, the air density is proportional to pressure: $\rho = pM/(RT)$, with $M = 0.029 \text{ kg/mol}$, $R = 8.314 \text{ J/(mol K)}$ and $T = 273 \text{ K}$. (a) Using $dp/dz = -\rho g$, show that $p = p_0 e^{-z/H}$, and find the scale height H . (b) Estimate the air pressure at the summit of Everest (8849 m) as a fraction of sea-level pressure. (c) Oxygen makes up 21% of air at every altitude. What is the partial pressure of oxygen at the summit, in mmHg? Comment on why climbers usually carry supplementary oxygen.

P10.18

Laplace's law and stability. (a) Two alveoli, of radii 0.050 mm and 0.10 mm, are connected to the same airway, and both have surface tension 0.070 N/m. Find the pressure needed to keep each one inflated. Which way would air flow between them, and what happens to the smaller one? (b) Explain how surfactant, whose surface tension falls as the surface area shrinks, stabilises the alveoli. (c) For an artery, the wall tension is $T = pr$. Use this to explain why an aneurysm, a local ballooning of the vessel wall, tends to keep growing once it starts.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

